[Total No. of Questions - 9] [Total No. of Printed Pages - 3]

Dec-25-0365

CSPC-515 (Data Warehouse and Data Mining)

B.Tech. 5th (NEP)

Time : 3 Hours **Max. Marks : 60**

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from section A, B, C and D. Section E is compulsory.

SECTION - A

1. (a) Define data warehouse with five features. Discuss its design principles. (6)

   (b) Discuss about multidimensional database, data mart and data cube. (6)

2. (a) Compare Star and Snowflake schema. (6)

   (b) Explain the following in OLAP:

   i. Roll up operation

   ii. Drill Down operation

   iii. Slice operation

   iv. Dice operation

   v. Pivot operation (6)

SECTION - B

3. Write and explain the algorithm for mining frequent item sets without candidate generation. Consider given database with nine transactions let min_sup = 30%. Find all frequent item sets.

TID	List of items_IDS
1	apple, ball, egg
2	ball, dice
3	ball, cake
4	apple, ball, dice
5	apple, cake
6	ball, cake
7	apple, cake
8	apple, ball, cake, egg
9	apple, ball, cake

(12)

4. (a) Explain the need for Data Preprocessing. (6)

   (b) Explain the concept of Data Discretization and Concept Hierarchy generation. (6)

SECTION - C

5. Differentiate between classification and regression. How is prediction related to both? (12)

6. For a two-class classification problem, with a Positive class P and a negative class N, we can describe the performance of the algorithm using the following terms: TP, FP, TN, and FN.

   i. What do each of these terms refer to?

   ii. Create a confusion matrix for these four terms.

   iii. Provide the formula for accuracy in terms of TP, TN, FP, and FN.

   iv. Provide the formula for precision and recall using TP, TN. FP and FN. (4×3=12)

[P.T.O.]

SECTION - D

7. Discuss the basic difference between the agglomerative and divisive hierarchical clustering algorithms and mention which type of hierarchical clustering algorithm is more commonly used, taking a suitable example. (12)

8. Consider five points {X_1, X_2, X_3, X_4, X_5} with the following coordinates as a two dimensional sample for clustering.

   X_1=(0.5, 2.5);

   X_2=(0,0);

   X_3 (1.5, 1);

   X_4 (5,1);

   X_5 (6, 2).

   Illustrate the K-means partitioning algorithms using the above data set just for one cycle. (12)

SECTION - E (Compulsory)

9. Answer following questions in brief:

   i. What is meta data?

   ii. What is correlation analysis?

   iii. How is the prediction is differed from classification?

   iv. What is the need of data preprocessing?

   v. What is knowledge in KDD process?

   vi. What is outlier? (6×2=12)